

Problem 2: Exploring Edge Impulse

1. Learning Blocks Available in the Free-Tier Edge Impulse

Supervised Learning – Classification

Used for predicting categories or labels.

Example Models

- MobileNet Image Classification
- Keyword Spotting
- Motion Classification

Supervised Learning – Regression

Used for predicting continuous values.

Example Models

- Temperature prediction
- Heart-rate estimation

Object Detection

Used to detect and locate objects in images.

Example Models

- FOMO
- MobileNet SSD

Unsupervised Learning / Anomaly Detection

Used to detect abnormal patterns.

Example Models

- K-Means Clustering
- Autoencoder anomaly detection

Transfer Learning / Self-Supervised Learning

Uses pretrained feature extractors.

Example Models

- MobileNet transfer learning
- Audio embeddings

2. Types of Layers in the Edge Impulse Classifier

Dense Layer

Performs final classification decisions.

Convolutional Layer

Extracts image or audio features.

Pooling Layer

Reduces computation and feature size.

Dropout Layer

Helps prevent overfitting.

Flatten Layer

Converts feature maps into vectors.

Activation Layers (ReLU, Softmax, Sigmoid)

Add nonlinearity and output probabilities.

Batch Normalization

Improves training stability.

Recurrent Layers (LSTM / GRU)

Processes sequential/time-series data.

3. Model Compression Options in Edge Impulse**Quantization**

- INT8
- Float16

Reduces model size and improves speed.

EON Compiler

Optimizes RAM and flash usage for embedded devices.

Pruning / Sparse Optimization

Removes unnecessary weights to reduce size.

TensorFlow Lite Conversion

Converts models into lightweight TFLite format.

Hardware-Specific Optimization

Example: CMSIS-NN for ARM Cortex-M processors

4. Synthetic Data Generation in Edge Impulse**Audio**

Methods:

- Noise injection

- Pitch shifting
- Time stretching

Images

Methods:

- Rotation
- Cropping
- Flipping
- Brightness adjustment

Sensor / Time-Series Data

Methods:

- Noise addition
- Time shifting
- Signal scaling

Importance of Synthetic Data

Synthetic data improves dataset diversity, reduces overfitting, and helps models generalize better to real-world conditions. It is especially useful when collecting large amounts of real data is difficult or expensive.